Book Review

Hillslope Hydrology and Stability

(Ning Lu and Jonathan W. Godt)

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Hillslope hydrology and stability forms an important addition to the literature on hydromechanical processes in hillslopes. The book is divided into five parts and ten constituent chapters. Part one provides an introduction and state-of-the-art. Important background on landslide terminology, classification, occurrence and socio-economic impacts are introduced with a summary of rainfall induced landslides. The authors then perhaps somewhat ambitiously cover hillslope geomorphology in one chapter predominantly describing the hillslope hydrologic cycle, topography, soil classification, hillslope hydrology and streamflow generation and mechanical processes in hillslopes.

The second part of the book concerns hillslope hydrology with chapters on steady and transient infiltration. Water movement mechanisms, Darcy's law and vertical flow in one layer and two layer soil systems are all discussed in detail. Hydrologic barriers including flat and dipping capillary barriers are introduced as well as hydrologic barriers due to heterogeneity. The governing equations for transient water flow are presented for saturated and unsaturated flow. One-dimensional transient flow is discussed with reference to Richards equation and the Green-Ampt and Srivastava and Yeh infiltration models. Numerical solutions for multi-dimensional problems are briefly outlined. A detailed account of transient flow patterns includes the factors controlling flow direction, a conceptual model for wetting and drying states and flow patterns under constant rainfall intensity. Discussion of transient hillslope flow patterns concludes with consideration of patterns after cessation of rainfall and resulting from transient rainfall.

The third part of the book moves from hydrology to a state-of-the-art discussion of total and effective stress in hillslopes. Total stresses are first defined and introductory stress-strain concepts and relationships stated. Analytical and graphical representation of the state of stress using the Mohr circle is outlined prior to discussion of force equilibrium equations. After consideration of two-dimensional elastostatics the authors present previous research on total stress distribution in hillslopes including Savage's two-dimensional analytical solutions. A section of the book is then devoted to presentation of total stress distributions in homogeneous and isotropic hillslopes inclined at varying slope angles. These stress distributions are based on finite element solutions and are given in stress contour plot and chart format for future use by the reader. The chapter on effective stress in soil begins with an introduction to Terzaghi's and Bishop's effective stress theories followed by Coleman's independent stress variable theory.

Perhaps one of the major differences and arguably controversial aspects of the book is the presentation and use of the first author's suction-stress theory. The authors state that the consideration of the two variables $(u_a - u_w)$ and $(\sigma - u_a)$ as stress variables is "flawed in that they are not stress quantities at the representative elemental volume and macroscopic levels". The authors go on to present the suction stress concept and the unified stress principle which forms the foundation for the much of the remainder of the treatment of slope stability in the book. After presentation of the unified equation for effective stress and discussion of its validity, suction stress profiles in hillslopes for single and multiple soil layers are provided. The reviewer found this part of the book, although presenting an interesting new approach, to perhaps lack in the balance required for a comprehensive text book that could be used by students and practicing engineers and geoscientists alike.

It is surprising that there is no reference in the book to the texts by Fredlund and coworkers (1993, 2012) on unsaturated soil mechanics in engineering practice given that most of the current state of practice and numerical analysis in geotechnical engineering are based on the approaches outlined in this text. The authors could have improved the book by providing more coverage of the alternatives to their theory and by showing clearly the differences that would be obtained in design practice using their proposed theory and the approach currently in use by most engineers. Although the reviewer believes a valuable alternative approach is outlined, the reader – whether student or practicing engineer or geoscientist – will need to consult other textbooks to have a balanced viewpoint and to be fully aware of the current state-of-the-art in everyday practice.

The fourth part of the book devotes two chapters to hillslope material properties. These chapters relate to the strength and hydromechanical properties of hillslope materials. The authors provide a useful introduction to shear strength due to frictional resistance and cohesion introducing concepts such as apparent cohesion, drained cohesion, cementation and capillary cohesion. A good review of the contribution of root reinforcement to the shear strength of rooted soils and the tensile strength of roots is provided along with the importance of spatial and temporal variations in root strength. After describing the shear strength of soils under varied drainage conditions the chapter on material strength concludes with the introduction of an equation for the unified shear strength criterion which incorporates the major contributing physical mechanisms to the shear strength of hillslope soils. The chapter on the hydro-mechanical properties of hillslope soils provides an introduction to the various methods of measuring suction and hydraulic conductivity emphasizing the derivation of the soil water retention curve (SWRC), the hydraulic conductivity function, (HCF) and the suction stress characteristic curve (SSCC). A major focus is on the transient release and imbibition method (TRIM), which is covered in detail with respect to the procedure, parameter identification, validation and application to different soils.

The final part of the book is concerned with hillslope stability with chapters focusing on failure surface and stress field based stability analysis, Classical methods of slope analysis are introduced beginning with the infinite slope and Culmann's finite slope stability model followed by description of the common methods of slices. The major focus of the chapter on failure surface based analysis is the analysis of landslides under steady infiltration and of shallow landslides induced by transient infiltration. The extension of classical methods to unsaturated conditions and the impact of infiltration rate and moisture rate are outlined. The stability of shallow landslides induced by transient infiltration is discussed with reference to varied hillslope soil types. The chapter concludes with two interesting case studies, one of a rainfall-induced shallow landslides and the other of a snowmelt-induced deeply seated landslide. The book concludes by introducing stress field based stability analysis and emphasizing the differences between failure based and finite element based hillslope analyses. The concept of a stress based scalar field or local factor of safety approach is introduced and a comparison between the classic methods of slices factor of safety and the local factor of safety methods presented for varying geometry slopes. The stress based local factor of safety approach is then extended to transient hillslope stability analysis and demonstrated through application to the two case studies introduced in the previous chapter.

Although the reviewer found this section of the book useful and interesting, an approach based on local factors of safety derived from elastic analyses alone is perhaps unusual given the published literature now existing on the merits of limit equilibrium analyses compared to numerical shear strength reduction methods based on elastoplastic theory. Such elastoplastic shear strength reduction methods, using a variety of constitutive models, are now used routinely in practice and available in a wide variety of continuum and discontinuum numerical codes.

In summary the book could have benefited by more discussion of the current practice and inclusion of more reference to case studies and more reference to the large body of field work on hillslope terrain analysis. This book has an emphasis on the theory pertaining to hydrology, unsaturated soil mechanics and slope analysis and will form a useful reference for graduate students and practicing geoscientists and engineers interested in these areas.

REFERENCES

Fredlund, D. G. and Rahardjo, H., 1993, *Soil Mechanics for Unsaturated Soils*. Hoboken, New Jersey.

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